

Segmentation categories

The following nine segmentation categories were proposed and agreed on for the services described in the offer. Furthermore, for the large-scale mining operations an excel-file with supplementary data has been compiled. The legend for this excel file is appended to this document.

Segmentation category	Abbreviation
artisanal and small-scale mining	asm
leaching heap	lh
mine facilities	mf
mine site area	ms
open-pit	op
processing plant	pp
stockyard	sy
tailings storage facility	tsf
waste rock dump	wr

The segmentation category, the *imaging-date* as well as the *imaging-system* are stored in the properties of the individual polygons in accordance with the following scheme:

```
{  
  "category":{"op": 0 ,"wr": 0 ,"tsf": 0 ,"lh": 0 ,"pp": 0 ,"mf": 0 ,"ms": 0 ,"sy": 0 ,"asm": 0},  
  "date": " year-month-day ",  
  "date2": " year-month-day ",  
  "system": " name of imaging system "  
}
```

Leaching heap (lh):

Heap leaching is an industrial mining process used to extract precious metals, copper, uranium, and other compounds from ore using a series of chemical reactions that absorb specific minerals and re-separate them after their division from other earth materials. Similar to in situ mining, heap leach mining differs in that it places ore on a liner, then adds the chemicals via drip systems to the ore, whereas in situ mining lacks these liners and pulls pregnant solutions up to obtain the minerals. Heap leaching is widely used in modern large-scale mining operations as it produces the desired concentrates at a lower cost compared to conventional processing methods such as flotation, agitation, and vat leaching ([Source](#)).

Leaching heaps (or leach pads or leach dumps) are typically stacked in layers on top of each other throughout the life of mine. Once one heap leach layer has been fully leached, another layer is stacked on top of it and leached again, resulting in a lower land consumption in comparison to building leach heaps next to each other.

Mine facilities (mf):

In the context of the annotation scheme that has been applied to the given contours of Chilean large-scale mining operation, “mine facilities” includes the following structures and buildings: Workshops, offices, living quarters, laboratories, canteen, parking areas, as well as all other structures that are not associated to the processing plant, the tailings storage facility, the waste dump, or the open pit of the mining operation.

Open-pit (op):

Contours marked with this annotation depict the outlines of the open pit(s). Thereby, not only active, but also inactive areas – which once belonged to an open pit and can still be clearly identified as a former open pit – are included. Some mining operations apply so-called “in-pit dumping”, where mined waste rock is dumped into sections of the open pit that have already been fully excavated and are not relevant for the extraction processes anymore. In those cases, the transition between open pit and waste rock dump (wr) is sometimes hard to delineate. In those cases, it was decided to label those corresponding parts of the open pit as waste rock dump (wr), as this represents the chronologically most recent function of the area. Labelling those areas also as “op” would not be favourable for this training data set, as these areas do not have the visual features of an “op”.

Waste rock dump (wr):

Waste rock is the bedrock that has been mined and transported out of the pit but does not have metal concentrations of economic interest. It is not fed to the processing plant. Instead, it is typically dumped onto large waste rock dumps located close to the open pit. In some cases, waste rock may still be suitable to be used as material for the construction of roads or other earthworks. In those cases, it may be temporarily dumped on very small waste dumps. Those areas have also been annotated as waste dumps.

Tailings storage facilities (tsf):

Tailings consist of ground rock and process effluents that are generated in a mine processing plant. Mechanical and chemical processes are used to extract the desired product from the run of the mine ore and produce a waste stream known as tailings. This process of product extraction is never 100% efficient, nor is it possible to reclaim all reusable and expended processing reagents and chemicals. The unrecoverable and uneconomic metals, minerals, chemicals, organics and process water are discharged, usually as slurry, to a final storage area commonly known as a Tailings Management Facility (TMF) or Tailings Storage Facility (TSF). ([Source](#))

There are different types and shapes of TSFs that also were encountered throughout the labelling of this data set. All of those types are typically fully/partially surrounded by a constructed dam, behind which the tailings are disposed.

Stockyard (sy):

Especially remote mining operations have large stockyards with spare parts for all sorts of equipment components, such as conveyor belts, metal structures, bearing, crusher cones, conveyor belt pulleys, truck tires. Those stockyard areas have been labelled with “sy”.

Processing Plant (pp):

Processing plants are fed with ore and aim to separate valuable minerals within the ore from the invaluable gangue minerals. Moreover, the valuable minerals are later concentrated to the final product of a processing plant, the concentrate. Typically, each processing plant has the following four primary operations: comminution (i.e. the reduction of the particle sizes to liberate the valuable minerals), sizing (i.e. the separation of particles sizes by screening or classification), concentration, and dewatering. In the applied labelling scheme, all processing plant-related facilities were labelled starting at the primary crusher (first stage of particle size reduction after blasting) and ending at the thickener ponds (dewatering of concentrate).

Artisanal and small-scale mining (asm):

There is no universal definition of artisanal and small-scale mining (ASM). The OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas (2013) defines ASM as "formal or informal mining operations with predominantly simplified forms of exploration, extraction, processing and transportation. ASM is normally low capital intensive and uses high labour-intensive technology."

In Ghana, the most common ASM mining method is hydraulic mining in combination with a sluice box. The basic setup consists of a pump system which is connected to two types of hoses: (a) squirting hose and (b) suction hose. High-pressure water jets loose the substrate and create a slurry that is a mixture of soil, gold particles and water. Then, the slurry is directed through the sluice boxes to separate the gold and the soil. The separation works because of the difference in density between the solids. The setup can be extended using excavators for topsoil removal.

OECD (2013), OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas: Second Edition, OECD Publishing.
<http://dx.doi.org/10.1787/9789264185050-en>.

Legend of supplementary excel file:

Column	Name	Description
D	Status	- Decommissioned → operations have already ceased - Operational → mine is still operating - Care & maintenance → mine is not actively operating, but regularly maintained and kept in such condition that allow for a mine restart in the future. - Landfill → those are contour IDs where no sign or information of a mining operation could be found during a detailed web search, and where all other information (location, size, images) indicate that it is a landfill. - N/A → those contour IDs could not be categorised. It is clearly not a mining operation. - Reclamation → those are decommissioned mining operations where reclamation has been carried out, so that there is only limited visual evidence of the mining operation in the past. - Exploration → evidence of former/current exploration activities (trenches, drill rig locations, test drifts), but no associated mining operation
E	Type	- OP: Open pit mining operation - UG: Underground mining operation - OP & UG: Parallel open pit and underground operation - OP → UG: Operation that has recently shifted from OP to UG, but still has substantial surface footprint of former OP operation
E	Processing Method	Description of key processing method(s) employed to process and concentrate the ore
J	ROM [Mt/a]	Indicated the annual run of mine (ROM) in million tons per annum. ROM includes the entire ore that is mined per year and fed to the processing plant. Extracted waste material is not part of the ROM.
K (N, Q, T)	Metal/Commodity #1 (#2, #3, #4)	Primary metal or commodity that is extracted at the mining operation
L (O, R, U)	Avg. grade #1 (#2, #3, #4)	Average ore grade of the mined ore
M (P, S, V)	Production #1 (#2, #3, #4)	Annual production (pure metal tonnage) in tons per annum
W, X, Y, Z, AA	Shareholder	Name and corresponding share of the main shareholding companies
AB, AC, AD, AE	Sources	URL of the corresponding sources